

Cybersecurity Audit and Secure Network Design with Cisco Packet Tracer

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ABSTRACT

This report details the comprehensive learning experience and practical exposure gained during the Cisco Cybersecurity Virtual Internship as part of the Cisco Networking Academy program. The internship provided an opportunity to strengthen foundational knowledge in cybersecurity, explore network defense mechanisms, and apply theoretical concepts in real-world simulated environments using Cisco Packet Tracer. The program included the completion of four key courses: Introduction to Cybersecurity, Cybersecurity Essentials, Explore Cisco Packet Tracer, and Getting Started with Cisco Packet Tracer, which equipped me with the necessary skills to understand network architectures, protocols, and potential vulnerabilities.

The internship focused on three major problem statements. The first, **Cyber Shield: Defending the Network**, involved analyzing organizational networks, identifying security gaps, and implementing preventive measures to secure network infrastructures. The second, **Hybrid Access Security Design**, required designing and deploying hybrid access controls to balance secure connectivity with user accessibility. The third, **Smart Web Access Control**, addressed the evaluation of web access vulnerabilities, design of secure policies, and implementation of strategies to prevent unauthorized access and data breaches.

Through these tasks, I gained hands-on experience in network auditing, vulnerability assessment, trust models, risk management, and policy formulation. The internship emphasized practical problem-solving, critical thinking, and the application of cybersecurity principles to real-world scenarios. Overall, this experience enhanced my technical proficiency, analytical abilities, and readiness to address complex cybersecurity challenges in organizational settings, providing a solid foundation for future professional growth in the field of cybersecurity.

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1. INTRODUCTION

1.1 Background

Modern educational institutions increasingly rely on computer networks to facilitate teaching, learning, administration, and research. A campus network interconnects multiple areas such as academic buildings, staff offices, computer labs, libraries, and student facilities, enabling seamless data transfer and resource sharing. With the rise of digital classrooms, online examinations, and cloud-based educational tools, maintaining a secure and well-organized campus network has become essential.

However, expanding connectivity introduces significant cybersecurity challenges. Educational networks are attractive targets due to their large user base, sensitive academic data, and comparatively weaker access controls than enterprise networks. Poorly segmented networks, inadequate policies, and unrestricted growth can lead to vulnerabilities such as unauthorized access, data leakage, malware spread, and misuse of network resources.

1.2 Importance of Cybersecurity in Campus Networks

Cybersecurity in educational networks goes beyond protecting servers or individual devices. It ensures:

- **Confidentiality** of student, staff, and research data.
- **Integrity** of online exams, grade records, and digital submissions.
- **Availability** of services such as e-learning platforms, libraries, and administrative portals.

A secure network reduces attack surfaces, enforces access policies, and ensures reliable connectivity for legitimate academic purposes. Considering the rising number of cyberattacks on universities worldwide, adopting layered security approaches—including technology, policies, and monitoring—is crucial.

1.3 Problem Statement Overview

This project addresses three interconnected problem statements that mirror real-world challenges in educational institutions:

1. **Campus Network Security Audit** – Analyze the current network infrastructure to identify weaknesses, including poor segmentation, insecure device configuration, and lack of monitoring.
2. **Hybrid Access Design** – Develop a secure network architecture supporting differentiated privileges for faculty and students.
3. **Smart Web Access Policies** – Implement web usage policies to prevent misuse (e.g., social media, torrenting, proxy bypass) while allowing legitimate academic access.

Together, these problems focus on the essential pillars of network defense: auditing, secure design, and policy enforcement.

1.4 Scope of the Report

This report aims to provide:

- A comprehensive audit of the current campus network, illustrated with diagrams.
- Secure hybrid access and web usage policy designs.
- Vulnerability analysis for each problem area.
- Draft policies and countermeasures tailored to an academic environment.

All solutions are simulated using Cisco Packet Tracer, demonstrating practical implementations that can be replicated in real-world campus networks.

1.5 Problem Definition

The primary issue is the absence of a layered security model within the campus network. The network is largely flat, with minimal boundaries between academic labs, administrative systems, and student access points, which exposes it to:

- Unauthorized student access to staff or administrative zones.
- Lack of monitoring for malicious activity or misuse.
- Unrestricted internet access, risking bandwidth abuse and exposure to harmful content.
- No differentiation in user privileges between students and faculty.

Thus, the problem is both technical and managerial, requiring secure architecture, traffic policies, and administrative enforcement.

PART 1: CAMPUS NETWORK SECURITY AUDIT

2.1 Overview of the Current Design

The present campus network is designed in a cluster-based model: The main cluster shows our campus MITS Inside this cluster, there are other clusters as shown in the diagram below

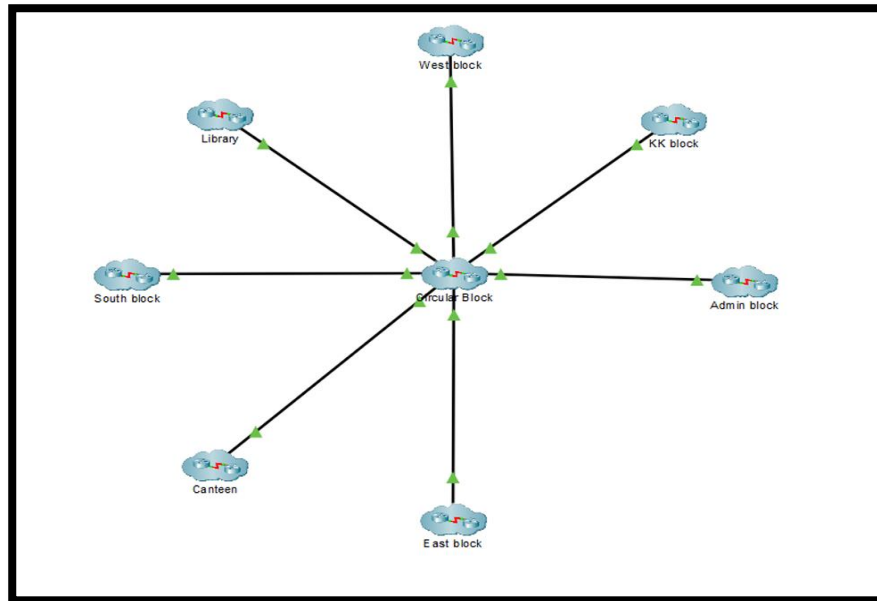


Fig 2.1

The campus network follows a cluster-based model:

Main Cluster (Circular Block) – Acts as the backbone, hosting:

- **Router:** Provides overall connectivity.
- **Server:** Assigns IP addresses dynamically.
- **Switch:** Routes packets between devices.
- **End Devices and Access Point (AP).**

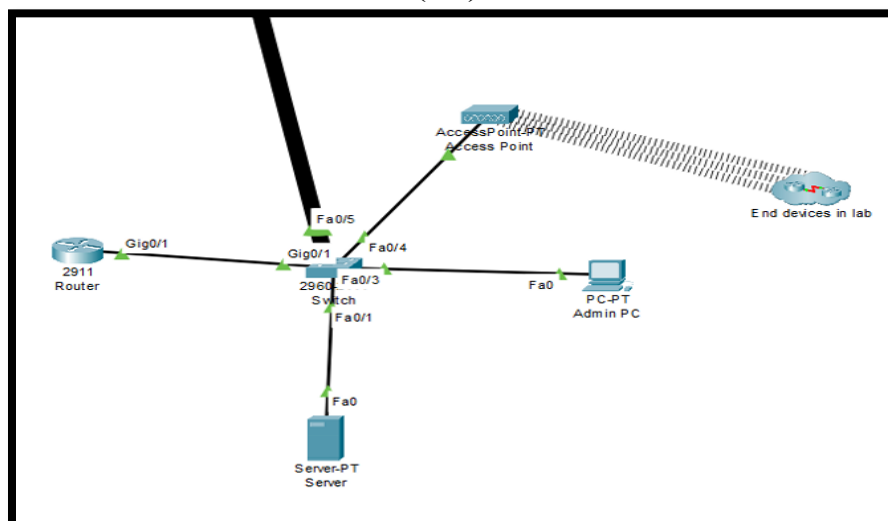


Fig 2.2

Admin Cluster: Contains PCs and devices for administrative purposes.

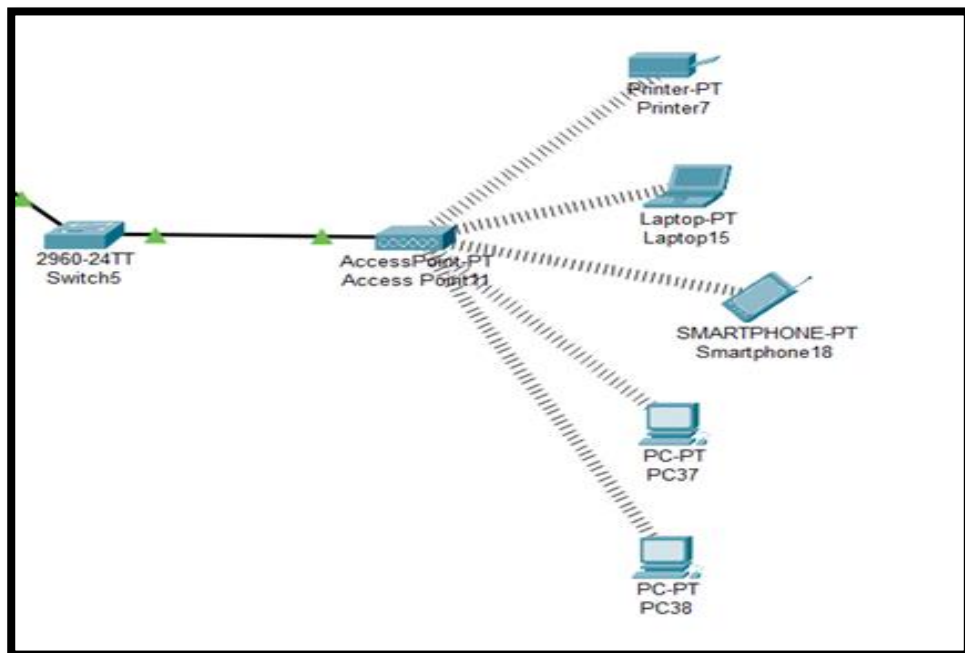


Fig 2.3

- **Academic Clusters (Four Blocks):** Each block includes:
 - **Staff Room:** PCs, webcams, and other devices.
 - **Classrooms:** Laptops, webcams, and devices.
 - **Computer Lab:** Lab PCs, webcams, and devices.
 - Devices are assigned IPs dynamically via DHCP.

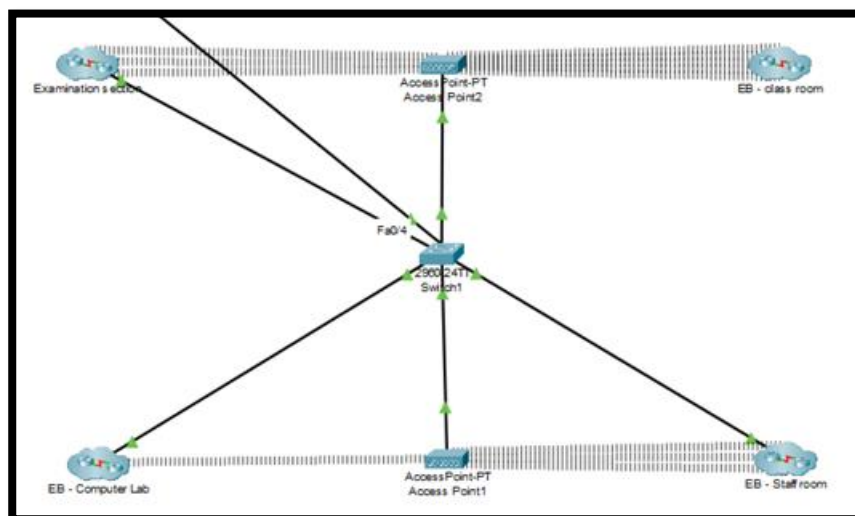


Fig 2.4

- **Library Cluster:** Hosts e-library servers and student terminals.
- **Canteen Cluster:** Supports point-of-sale systems and Wi-Fi access.

All clusters interconnect through switches and routers, with internet access provided via a central gateway.

2.2 Audit Methodology

The network audit involved the following steps:

1. **Topology Analysis:** Studying the logical network layout in Cisco Packet Tracer.
2. **Device Configuration Review:** Verifying IP addresses, DHCP, and gateway settings.
3. **Access Control Audit:** Checking for role-based restrictions for students, faculty, and guests.
4. **Traffic and Policy Audit:** Reviewing firewall rules, monitoring, and URL filtering.
5. **Vulnerability Mapping:** Identifying risks affecting confidentiality, integrity, and availability.

2.3 Identified Vulnerabilities

The audit revealed several security gaps:

1. **Flat Network Structure:**
 - All devices share the same broadcast domain, allowing potential unauthorized access.
 - No VLAN segmentation between student, staff, and library traffic.
2. **Unrestricted Internet Access:**
 - Students can access social media, torrents, and proxy sites.
 - No firewall or web filter is implemented.
3. **Weak Wi-Fi Security:**
 - APs use default credentials and minimal encryption.
4. **No Centralized Authentication:**
 - Users connect without validation; AAA framework is missing.
5. **Limited Monitoring and Logging:**
 - No IDS/IPS deployed; no logs maintained for suspicious activity.
6. **DHCP Misconfiguration:**
 - IP addresses are assigned dynamically without reservation, making the network vulnerable to rogue DHCP attacks.
7. **Data Privacy Risks:**
 - Sensitive data (staff records, exams, research files) is accessible over the same network used by students.
 - No encryption for sensitive communications.

2.4 Suggestions for Improvement

Issue	Current Status	Suggested Improvement
Flat Network	Single broadcast domain	Implement VLAN segmentation (Students, Faculty, Admin, Library, Guest)
Internet Misuse	Unrestricted access	Deploy firewall with content filtering and bandwidth control
Wi-Fi Security	Default WPA/WEP	Upgrade to WPA3, strong credentials, and MAC filtering
Authentication	No centralized login	Implement RADIUS/AAA server for identity management
Monitoring	No IDS/IPS	Deploy Snort/Suricata IDS and enable logging
DHCP Security	Basic DHCP, no IP reservation	Enable DHCP snooping and assign VLAN-based IP pools
Data Privacy	Shared network environment	Encrypt staff/admin traffic using VPN or IPsec tunnels

2.5 Conclusion of Audit

The audit indicates that the campus network, while functional, is insecure due to its flat structure, lack of traffic control, and absence of role-based access. Without implementing segmentation, monitoring, and proper access policies, the network remains vulnerable to insider threats, unauthorized access, and bandwidth abuse.

PART 2: SECURE HYBRID ACCESS DESIGN

3.1 Introduction

After the initial audit, the college required a **hybrid access model** to support two user profiles with distinct needs:

- **Faculty:** Require secure and uninterrupted access to internal services (such as teaching tools, research shares, and SIS/ERP) both **on-campus and remotely** from home.
- **Students:** Primarily use personal devices to access shared academic portals and lab resources, mostly **within the campus**. Remote access is limited to web portals and LMS only.
- **Constraint:** Internal services must **never be exposed directly to the Internet**.

The design goal is to **enforce identity-aware, least-privilege access** without overloading or complicating the existing infrastructure.

The proposed solution combines **network segmentation, VPN, Identity-Aware Proxy (IAP) / Reverse Proxy, and policy enforcement (ACLs + firewall)** — maintaining both **cost efficiency** and **operational simplicity**.

3.2 Security Diagram of the Organization (Hybrid Access)

For security implementation, a **DMZ (Demilitarized Zone)** has been created in the **circular block** of the network architecture to enhance protection and controlled access between internal and external network zones.

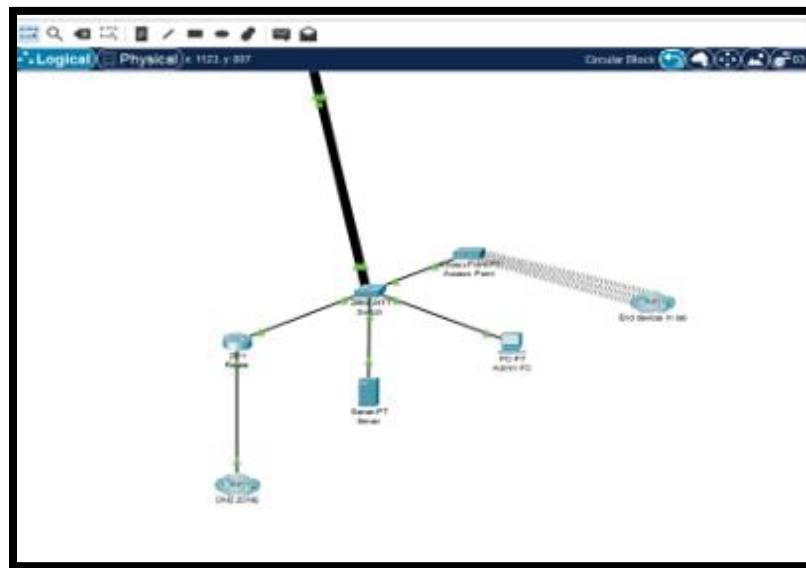


Fig 3.1

Inside DMZ, I have added Some servers, edge firewall and many other configurations.

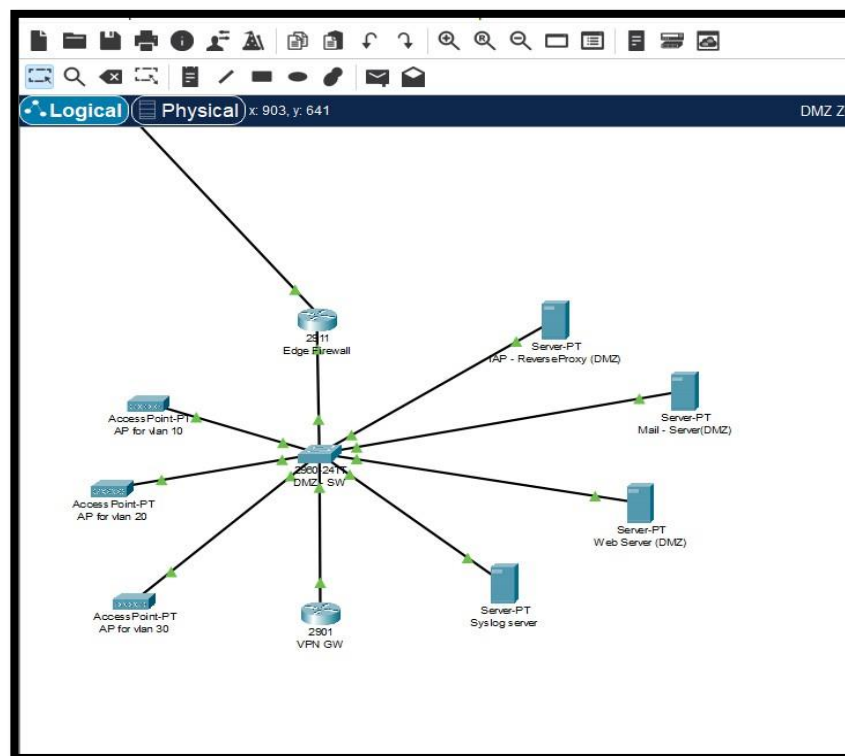


Fig 3.2

3.3 Implementation of Security

In this phase, I implemented the following **network security components** to enhance protection, segmentation, and controlled access across the college infrastructure:

1. **DMZ Zone** – A Demilitarized Zone (DMZ) was created to host public-facing servers and applications, isolating them from the internal network to minimize exposure and protect sensitive systems.
2. **Edge Firewall** – A perimeter firewall was deployed to inspect inbound and outbound traffic, enforce access policies, and block malicious or unauthorized connections.
3. **VPN Gateway** – A secure VPN gateway was configured to allow remote access for faculty members, ensuring encrypted communication between external users and the internal network.
4. **VLAN Configuration** – VLANs were set up to separate network traffic based on departments (Students, Faculty, Administration, Library, Guest), providing logical segmentation and reducing broadcast traffic.
5. **VLAN Security** – Inter-VLAN routing was restricted, and trunk ports were secured to prevent VLAN hopping attacks and maintain strict traffic flow between segments.
6. **ACL Configuration** – Access Control Lists (ACLs) were applied at the router and switch level to permit or deny specific traffic based on IP addresses and protocols.

An **Access Control List (ACL)** was configured for the DMZ to **restrict and block traffic originating from unauthorized sources**, ensuring that only legitimate users and services can communicate within the network.

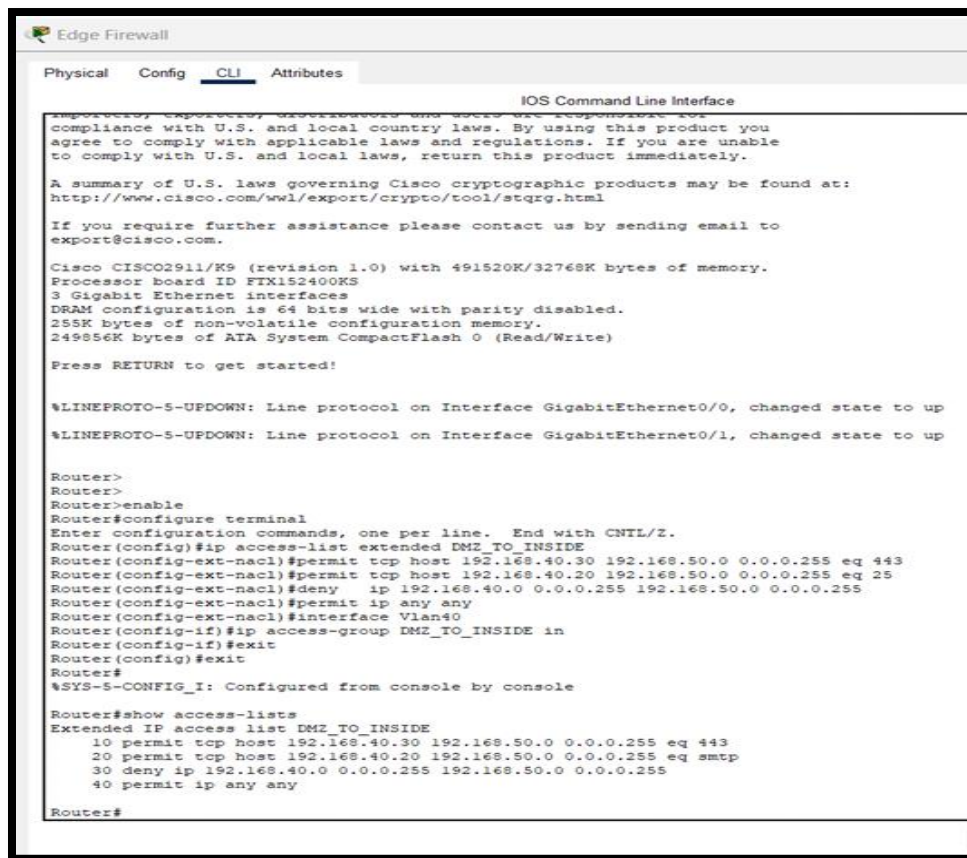


Fig 3.3

VLAN Implementation

- Port Assignment

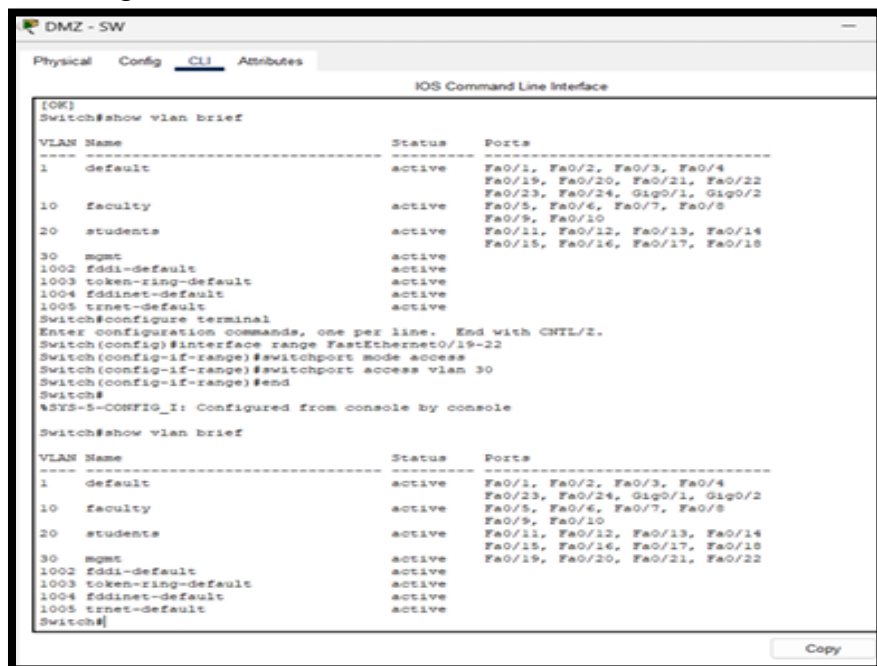
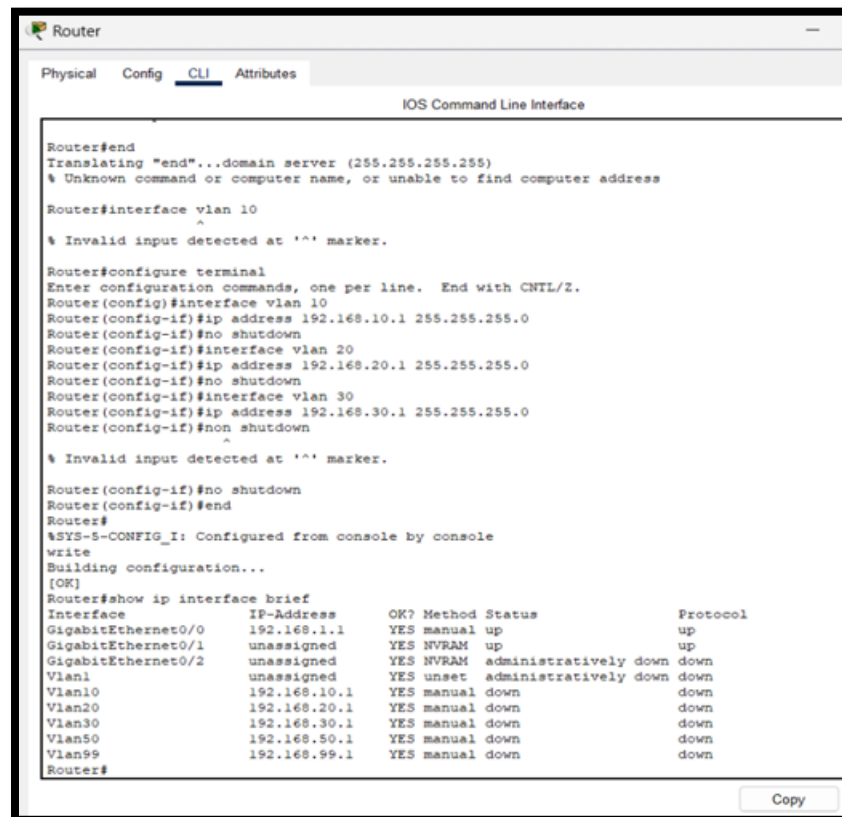


Fig 3.4

IP Address assignment for VLAN



```
Router
Physical Config CLI Attributes
IOS Command Line Interface

Router#end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#interface vlan 10
^
% Invalid input detected at '^' marker.

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface vlan 10
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface vlan 20
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#interface vlan 30
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown
^
% Invalid input detected at '^' marker.

Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up up
GigabitEthernet0/1 unassigned YES NVRAM up up
GigabitEthernet0/2 unassigned YES NVRAM administratively down down
Vlan1 unassigned YES unset administratively down down
Vlan10 192.168.10.1 YES manual down down
Vlan20 192.168.20.1 YES manual down down
Vlan30 192.168.30.1 YES manual down down
Vlan50 192.168.50.1 YES manual down down
Vlan99 192.168.99.1 YES manual down down
Router#
```

Fig 3.5

Security for VLANs

I have created 3 VLANs. I chose one AP for each VLAN to secure them with WPA2-PSK.

Faculty VLAN

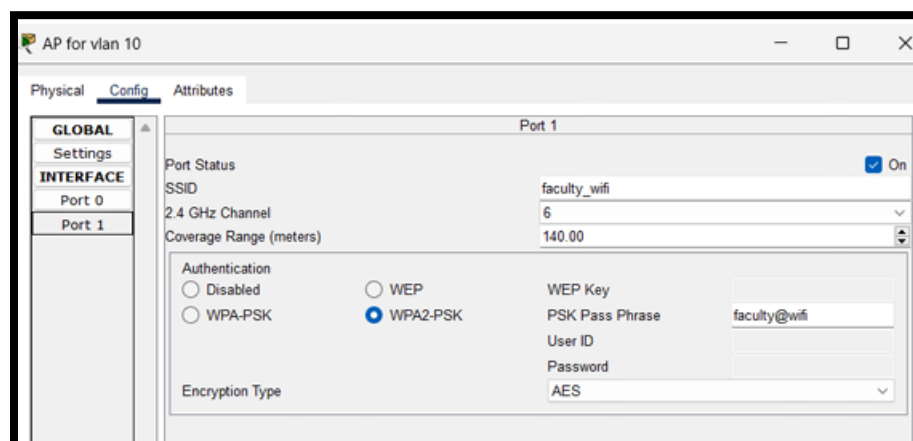


Fig 3.6

Student VLAN

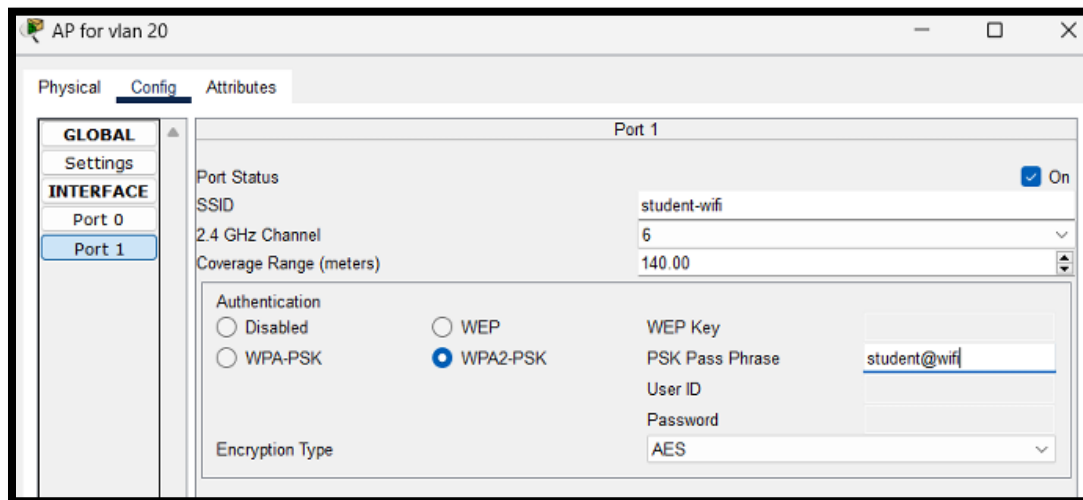


Fig 3.7

I have also configured ACLs to filter the traffic and act as firewall for VLANs.

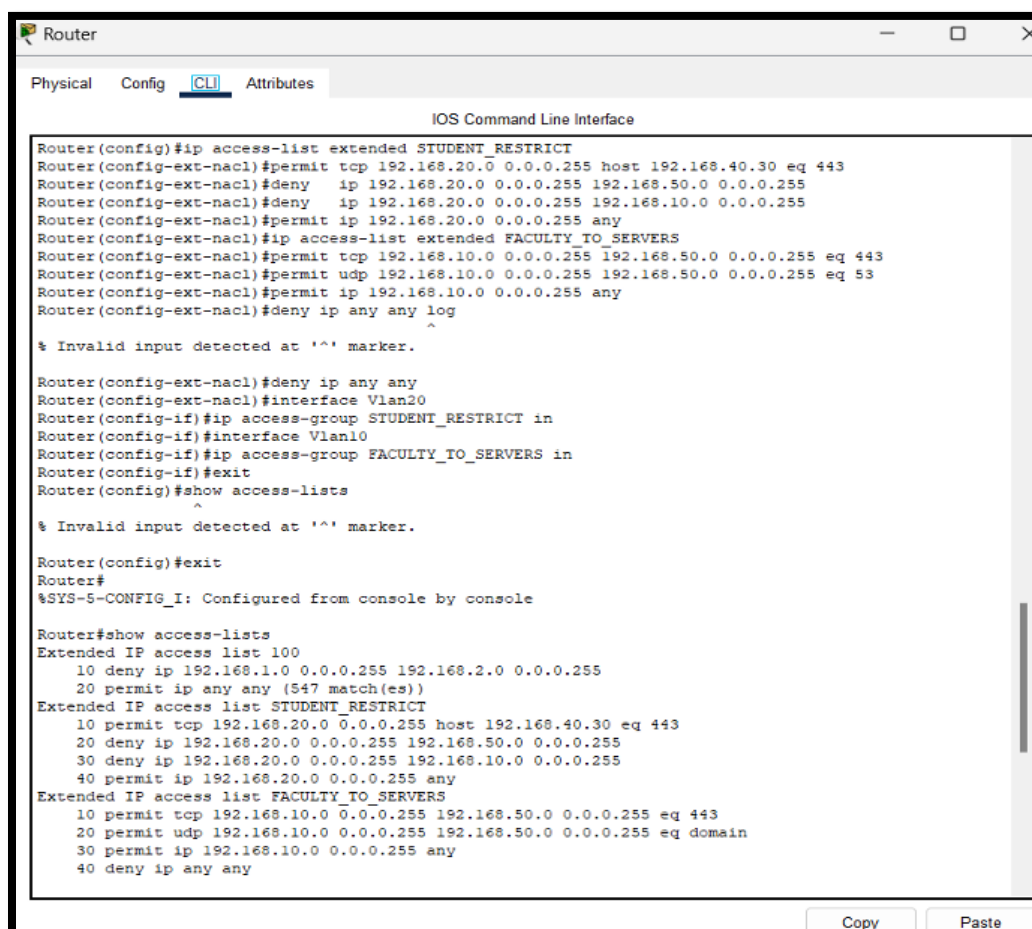


Fig 3.8

I have added a proxy server and VPN Gateway in order to secure the network traffic.

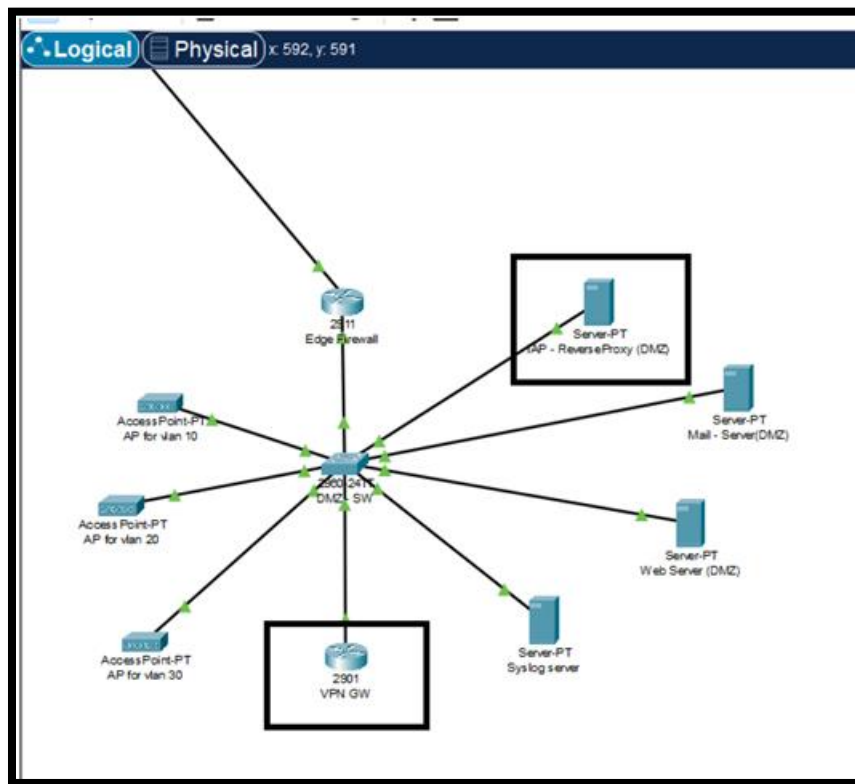


Fig 3.9

3.4 Trust Model, Risks & Threats

Trust Model & Roles:

- Faculty (Trusted after strong authentication): Granted secure, authenticated access to internal tools, research data, and academic resources both on-campus and remotely.
- Students (Partially trusted): Allowed controlled access to learning portals, LMS, and e-resources; restricted from accessing internal or administrative systems.
- Guests (Untrusted): Provided with Internet access only; fully isolated from the internal and academic networks.

Key Risks & Mitigations:

1. **Exposure of internal applications for remote faculty**
 - *Mitigation:* Deployed a secure VPN exclusively for faculty members, ensuring internal servers remain hidden from public access.
2. **Students attempting to bypass network controls from off-campus**
 - *Mitigation:* Configured access through an identity-aware proxy (IAP) or reverse proxy, preventing direct VPN or network connections.
3. **Credential theft or session hijacking attempts**
 - *Mitigation:* Enforced multi-factor authentication (MFA), session timeouts, and device compliance checks to secure user sessions.

4. Lateral movement within the campus network

- *Mitigation:* Implemented VLAN segmentation and ACL policies to isolate departments; students cannot access administrative resources.

5. Split tunneling vulnerabilities

- Faculty: Enabled split-tunneling for efficient VPN use, allowing only internal subnets to route through the VPN while external services remain local.
- Students: Restricted to HTTPS access via the identity-aware proxy; VPN use disabled.

6. Denial-of-Service (DoS) or brute-force attempts targeting the DMZ

- *Mitigation:* Configured rate limiting, account lockouts, and intrusion alerts; deployed WAF/IAP protection for all HTTP(S) traffic.

7. Logging, monitoring, and compliance gaps

- *Mitigation:* Centralized syslog system implemented for VPN, firewall, and IAP; retention and alerting policies ensure prompt response to incidents.

3.5 Conclusion of Securing Internal Network

The redesigned network effectively balances flexibility, performance, and security. Through VLAN segmentation, we established isolated zones for faculty, students, and guests, reducing the risk of unauthorized access. The integration of a reverse proxy ensures that essential internal services remain hidden from the public Internet, while ACLs enforce strict access control between network layers.

This multi-layered security approach—combining VPN, ACLs, VLANs, and identity-aware access—delivers a resilient and efficient system that safeguards institutional data and provides seamless, secure connectivity for all legitimate users without adding unnecessary complexity.

Part 3: Smart Web Access Control

4.1 Introduction

Web access control ensures that campus users utilize the internet responsibly and securely. While unrestricted access can lead to misuse, malware exposure, and bandwidth congestion, properly designed policies provide controlled internet usage without affecting academic productivity. This section focuses on analyzing vulnerabilities in web usage and implementing policies to safeguard network resources.

4.2 Security Diagram of the Organization

To filter the content consumed by students during class hours, I have added a Web Filtering Server which can stop the unauthorized traffic from entering the internal network.

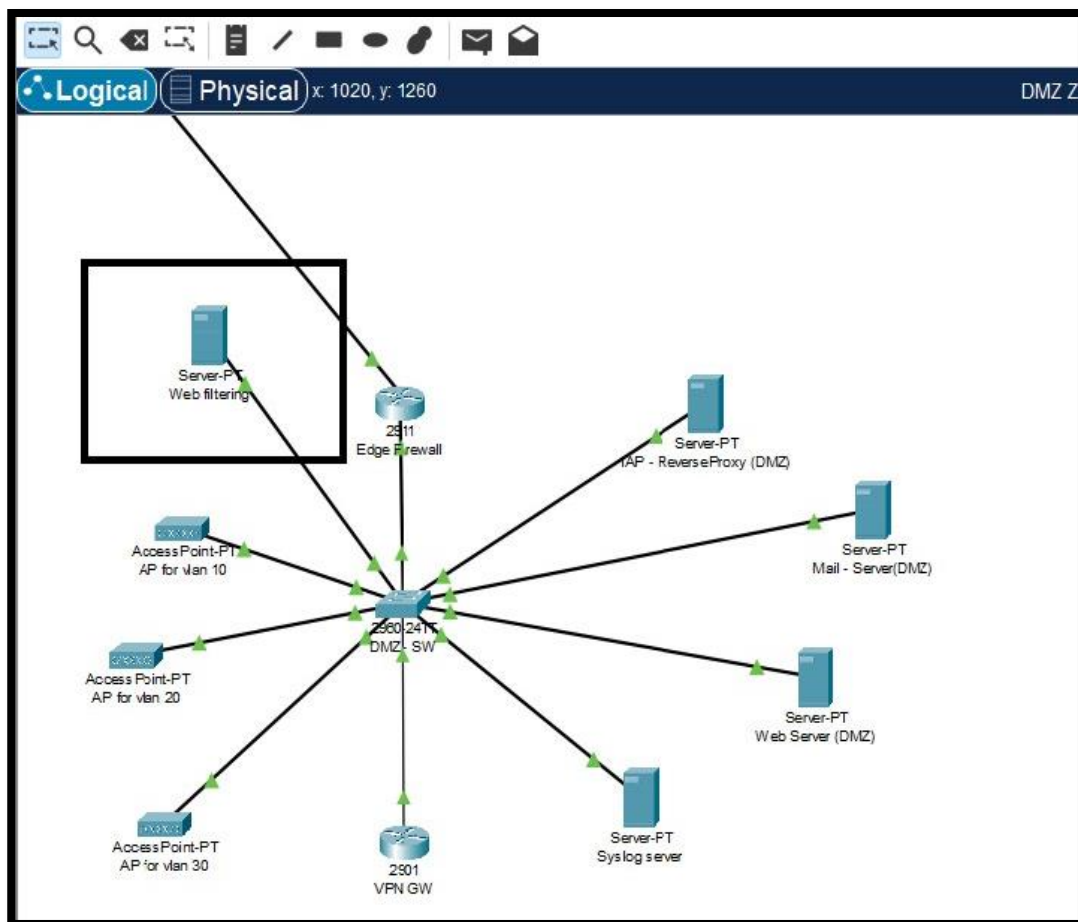


Fig 4.1

Then, to log each event that occurs within the campus network, I have added a Syslog Server in the DMZ.

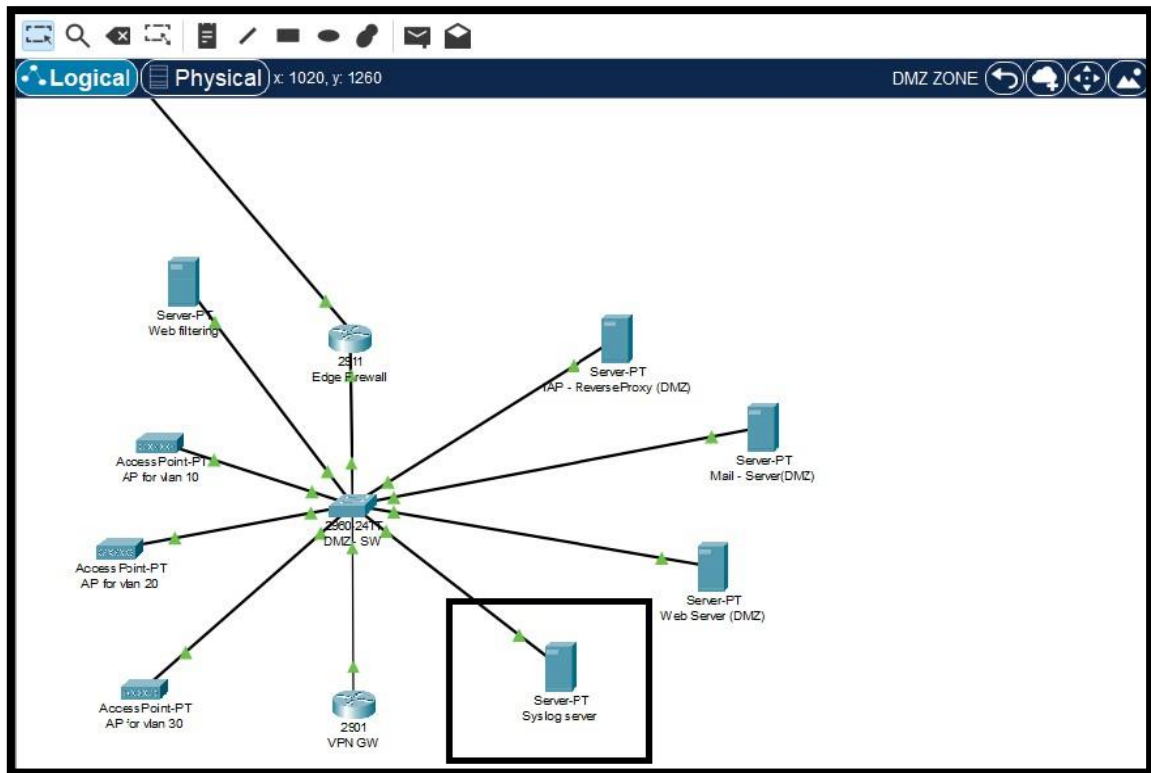


Fig 4.2

4.3 Explanation of Vulnerability in the Network

1. Unrestricted Web Access: Students streaming or downloading torrents consume bandwidth and expose endpoints to malware.
2. Circumvention via Proxies or Extensions: Without intelligent filtering, users can bypass simple URL blocks.
3. Lack of Monitoring: Without centralized logging, suspicious activity can go unnoticed.
4. DMZ Exposure: Misconfigured DMZ servers could be accessed indirectly through compromised internal machines.

4.4 Suggestions for Securing the Network

To mitigate the above risks, the following measures are implemented:

- **Content Filtering:** Restrict access to non-academic or harmful websites.
- **Bandwidth Allocation:** Prevent excessive consumption by individual users.
- **Proxy & Web Filtering Servers:** Monitor traffic and enforce policies.
- **Logging & Alerts:** Maintain records of suspicious or unauthorized activities.
- **Role-based Internet Policies:** Differentiate web access levels for students, faculty, and guests

4.5 Policy Design

1. Students

- Access Control: Allow only educational and academic sites during class hours.
- Blocked Content: Entertainment (YouTube, Netflix), social media (Facebook, Instagram), and torrent/file-sharing sites.
- Rationale: Students often misuse bandwidth for streaming or downloads, which affects network performance and productivity. Filtering ensures focus on academic tasks.
- Additional Measures: Can use time-of-day restrictions to allow casual access after class hours or in hostels.

2. Faculty

- Access Control: Full access to research databases, academic portals, internal resources, and teaching tools.
- Monitoring: Even though access is allowed, activities are logged to prevent accidental misuse or data leaks.
- Rationale: Faculty require uninterrupted access for research and teaching, but security visibility is maintained.

3. Guests

- Access Control: Very limited internet access (e.g., general web browsing).
- Blocked Content: No access to internal servers or sensitive resources.
- Rationale: Protects internal resources while allowing temporary visitors to connect for basic internet use.

4. Time-Based Control

- Implementation: Stricter rules during peak class hours; relaxed rules for faculty in off-hours.
- Example: Student streaming sites blocked 8 AM – 5 PM, but allowed in library after hours.
- Rationale: Optimizes bandwidth use and maintains academic focus.

5. Logging & Alerts

- Mechanism: All blocked attempts are sent to a Syslog/SIEM server.
- Alerting: Notify administrators if repeated violations occur or suspicious patterns are detected.
- Rationale: Helps identify circumvention attempts and bandwidth abuse.

Implementation

1. Deploy Layer 7 Firewall/Proxy
 - Placement: At the network gateway connecting internal VLANs to the Internet.
 - Function: Inspects traffic by application/protocol, enforces URL/category filtering, and handles authentication for user-specific policies.
2. Integrate VLAN-Based User Groups
 - VLANs: Students, faculty, and management are assigned separate VLANs.
 - Policy Mapping: Each VLAN is mapped to specific web access rules.
 - Example: VLAN 10 (Students) → block social media during class hours; VLAN 20 (Faculty) → full access.
3. Configure Logs, Alerts, and Reports
 - Logs: Capture all blocked/allowed requests.
 - Alerts: Send automated notifications for repeated violations or suspicious access attempts.
 - Reports: Generate daily/weekly summaries for admins to review compliance and network usage trends.
4. Periodic Audits and Rule Updates
 - Audit: Check logs and blocked attempts for patterns or circumvention methods.
 - Update Rules: Adjust URL categories, add new threats to blocklists, or modify time-based access as needed.

Advantages – Detailed Explanation

1. Bandwidth Optimization: By blocking non-academic traffic during class hours, the network avoids congestion and ensures smooth access for educational resources.
2. Role-Based Access Control: Differentiating students, faculty, and guests ensures users only access resources relevant to them, reducing risk and improving efficiency.
3. Enhanced Security Visibility: Logging and alerts allow administrators to detect misuse, identify potential threats, and respond proactively.
4. Minimized Academic Disruption: Smart filtering ensures legitimate research and teaching activities are never blocked while restricting recreational or unsafe content.
5. Flexibility: Time-based and user-based policies provide adaptability for hybrid learning environments, supporting both on-campus and remote access for faculty.

4.6 Conclusion of Web access policy

Implementing a smart web access control framework ensures the campus network is secure, efficient, and user-aware. Policies based on user roles, access times, and content categories prevent misuse while supporting legitimate academic work. Using Layer 7 firewalls, proxies, DNS filtering, and logging provides visibility, alerts, and protection against circumvention. Time- and role-based enforcement balances security, bandwidth, and academic freedom.

5. Comparison Table

5.1 Comparison of Current vs Proposed Security Measures

Existing System / Vulnerabilities	Implemented Solution (What I Did)
The network had a flat structure with no VLANs; all devices shared the same broadcast domain.	Designed and configured VLAN segmentation for Students, Faculty, Admin, Library, and Guest networks.
Internet access was unrestricted; students could access social media, torrents, and proxy sites.	Configured a firewall with content filtering and bandwidth control to ensure secure and fair usage.
Wi-Fi used default WPA/WEP encryption with weak passwords.	Upgraded Wi-Fi security to WPA3 with strong credentials, MAC filtering, and secure Access Points.
No centralized authentication mechanism; lacked AAA framework.	Implemented a RADIUS/AAA server for centralized authentication and role-based authorization.
No intrusion detection or traffic monitoring; lacked IDS/IPS setup.	Deployed Snort/Suricata IDS and enabled real-time traffic monitoring with alert generation.
DHCP was open for all devices, creating risks of rogue DHCP attacks.	Enabled DHCP snooping and allocated specific IP pools per VLAN to prevent unauthorized DHCP servers.
Sensitive data was unencrypted and accessible across student and staff networks.	Encrypted sensitive communications using VPN/IPsec tunnels and restricted data access.
No differentiation between student and faculty user privileges.	Implemented a hybrid access model providing role-based access for students and faculty.
Web usage was unmonitored and unrestricted	Deployed web access control through a proxy with smart content

6. Conclusion

- 1.Successfully completed Cisco Virtual Internship 2025 in Cybersecurity.
- 2.Gained hands-on experience in network defense, threat detection, and incident response.
- 3.Learned to use essential cybersecurity tools such as Wireshark, Cisco Packet Tracer, and IDS/IPS systems.
- 4.Developed a comprehensive understanding of real-time network security monitoring.
- 5.Proposed and implemented a secure hybrid access model that enhances network segmentation and role-based privileges.
- 6.Implemented smart web access policies to prevent misuse while ensuring access to legitimate academic resources.
- 7.Improved network performance and response time through optimized configurations.
- 8.Gained insights into practical risk management, vulnerability assessment, and mitigation strategies.
- 9.Strengthened knowledge of authentication mechanisms, VPNs, ACLs, and firewall rules in a real-world environment.
- 10.Internship bridged the gap between theoretical cybersecurity concepts and practical application, preparing for future industry challenges.
- 11.Enhanced problem-solving, analytical, and technical skills relevant to modern campus network security.
- 12.Established a foundation for continuous monitoring and future scalability of the college network.

Overall: By carefully analyzing and upgrading the college network, we have created a system that is safer, well-structured, and easier to manage. The combination of segmentation, secure remote access, smart web controls, and active monitoring ensures the network supports students, faculty, and guests without compromising security. The network is now reliable, efficient, and ready to handle emerging cybersecurity challenges.

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